Unit 5: International Trade and Finance

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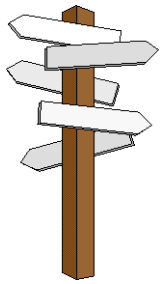
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Must Read Before You Begin

So far, we have only studied microeconomics and focused on the domestic sector of an economy. In this unit, we will turn our attention to macroeconomics and the external sector by looking at international trade.

This unit will first discuss the concept of comparative advantage and explain how countries benefit from specialisation and trading with each other. Then we will address the issue of protectionism. You may skip the part on comparative advantage because it is not required in this subject. But the concept will help you understand why a country exports certain goods while imports other goods.

Since international trade involves flows of currencies between countries, this unit has also included the foreign exchange market although it will not be covered in the lecture. The knowledge of how the foreign exchange market operates will prove to be very useful for your decision making on your job and for your personal investment in the future. It can be easily understood by applying the demand-supply analysis with which you should be very familiar by now. We will study how the equilibrium exchange rate of a currency is determined by the interaction between demand and supply of this currency under the flexible exchange rate system and the fixed exchange rate system. Again, you may skip this part if you cannot spare the time.

Please note that only Topic 5.2 is required in the examination of AF3625.

Topics

There are four main topics in Unit 5.

Topic 5.1: Absolute Advantage versus Comparative Advantage

A country has an absolute advantage in the production of a product when it uses fewer resources to produce that product than other countries do. However, you will see in this topic that absolute advantage is not a reason for the existence of international trade. Instead, comparative advantage provides an economic basis for trade – whether to trade or not depends on the existence of comparative advantage.

Topic 5.2: Gain from Trade and Trade Barriers

Based on the comparative advantage theory in Topic 5.1, both importing countries and exporting countries benefit from trade. However, in reality we observe various kinds of trade barriers that reduce external trade. This topic will examine the two most common forms of trade barriers – tariff and quota, and explain why these trade barriers are imposed.

Topic 5.3: Demand and Supply of Foreign Exchange

International trade involves the flow of various currencies in the foreign exchange markets. In this topic, we will first derive the demand curve and the supply curve of foreign exchanges. We will then discuss the factors that shift the demand curve and supply curve of foreign exchanges.

Topic 5.4: Determination of Equilibrium Exchange Rate

After deriving the demand curve and the supply curve of foreign exchanges in Topic 5.3, this topic will put them together to determine the equilibrium exchange rate under two different exchange rate systems – flexible exchange rate and fixed exchange rate. We will see how a government maintains a fixed exchange rate system by the use of foreign exchange reserves. In the final section of this topic, we will also take a look at the linked exchange rate system of the Hong Kong dollar.

Your Objectives

When you have worked through this unit you should be able to complete the following objectives.

1. Explain how the theory of comparative advantage determines the pattern of trade.
2. Explain why free trade is beneficial to the countries involved.
3. Analyse how protectionism and trade restrictions affect the parties involved.
4. Identify the determinants of demand and supply of a currency.
5. Analyse how government policies and other economic factors affect the exchange rate of a currency.

Self Exercises

There are four (4) self exercises in this unit.

Let us move on to the first topic in Unit 5.

5.1: Absolute Advantage versus Comparative Advantage

In this topic we will first define absolute advantage and comparative advantage. With the help of a hypothetical example, we will then explain the existence of international trade and study how the direction of trade is determined by comparative advantage instead of absolute advantage. As already mentioned earlier, if you cannot spare the time, you can skip this topic and move on to Topic 5.2.

5.1.1: Definitions

In this section, we will define what absolute advantage and comparative advantage are.

Absolute Advantage

A country has an absolute advantage in producing a product when it uses fewer resources to produce that product than other countries.

Suppose both Country A and Country B produce clothing, but Country A's labour is more productive. Country A will produce more clothing per unit of labour than Country B. Therefore, Country A enjoys an absolute advantage over Country B in the production of clothing.

Comparative Advantage

A country has a comparative advantage in producing a product when it can produce that product at a lower cost in terms of other goods than what other countries can do. In other words, a comparative advantage in producing a product means a **lower opportunity cost** of producing that product than other countries. (The concept of opportunity cost has been explained in Unit 1.)

Suppose both Country A and Country B produce clothing and Country A has an absolute advantage over Country B in doing that. However, having an absolute advantage does not necessarily mean having a comparative advantage.

Suppose both Country A and Country B also produce shoes. Further suppose that Country A and Country B each has a fixed amount of labour for producing shoes and clothing. Country A and Country B must each allocate its limited resources between the production of shoes and clothing. To produce more shoes, either country has to transfer resources from the production of clothing to the production of shoes. To produce more clothing, they have to do the reverse. This is exactly the concept of opportunity cost! The cost of shoes in each country can be measured in terms of clothing, and the cost of clothing can be measured in terms of shoes.

It will be easier to explain the concepts of absolute advantage and comparative advantage with a simple numerical example. Suppose each country is endowed with 24,000 units of labour, which is the only resource needed in the production. The costs of producing shoes and clothing in both countries are summarised in Table 5.1.

Table 5.1
Resource Requirement for Producing Shoes and Clothing

	1 pair of shoes	**1 piece of clothing**
Country A	3 units of labour	6 units of labour
Country B	8 units of labour	8 units of labour

From the above table, we find that Country A uses fewer units of labour to produce either 1 pair of shoes or 1 piece of clothing than Country B does. That is to say, Country A has an absolute advantage over Country B in the production of both shoes and clothing.

Does Country A have a comparative advantage in the production of both goods as well? To answer the question, we need to examine the opportunity cost of producing the two goods. According to the information given in Table 5.1, in Country A it requires twice the amount of resource to produce one piece of clothing compared to one pair of shoes. That is, to produce an additional piece of clothing, Country A must give up 2 pairs of shoes. Therefore, a piece of clothing has an opportunity cost of 2 pairs of shoes. At the same time, producing one piece of clothing in Country B requires the sacrifice of only 1 pair of shoes. Even though Country A has an **absolute advantage** over Country B in the production of clothing, Country B has a **comparative advantage** over Country A in the production of clothing.

In the next section, we will explain with the same example that even though Country A has an absolute advantage in both clothing and shoes, the possession of comparative advantage in different goods by the two countries provides a basis for beneficial trade between the two countries.

5.1.2: Comparative Advantage and Direction of Trade

In this section, using the example introduced in the previous section, we will illustrate how comparative advantage generates beneficial trade between two countries. That is, each country will be better off by specialising in the production of the product in which it has comparative advantage and trading with each other.

Hypothetical Example

Let us continue with the example of Country A and Country B described in the last section. Based on the information given in Table 5.1, is it worthwhile for Country A to trade with Country B when Country A can produce both goods more cheaply than Country B? The answer is YES!

Using the information in Table 5.1, we can derive the opportunity costs of producing shoes and clothing in Country A and Country B. Table 5.2 summarises the results.

Table 5.2
Opportunity Cost of Producing Shoes and Clothing

Opportunity Cost	1 pair of shoes	1 piece of clothing
Country A	0.5 piece of clothing	2 pairs of shoes
Country B	1 piece of clothing	1 pair of shoes

The above table shows that Country A has a comparative advantage in producing shoes and Country B has a comparative advantage in producing clothing.

Now, we will show that both Country A and Country B will benefit from trade if:

- **Country A specialises in the production of shoes, and**
- **Country B specialises in the production of clothing.**

In the remaining part of this section, we will consider **Case 1**: the two countries do not trade with each other and **Case 2**: the two countries trade with each other. We will then compare the total output in Case 1 with that in Case 2 to see if the two countries benefit from trade.

Case 1: No Trade (Autarky)

Suppose, without trade, each country divides its own resources equally between the production of shoes and clothing. Table 5.3 shows the production in Country A and Country B.

Table 5.3
Production of Country A and Country B without Trade

	Production of Shoes	Production of Clothing
Country A	12,000/3 = 4,000 pairs	12,000/6 = 2,000 pieces
Country B	12,000/8 = 1,500 pairs	12,000/8 = 1,500 pieces
Total	**5,500 pairs**	**3,500 pieces**

The production of shoes and clothing can also be illustrated by the production possibility frontiers for Country A and Country B in Figure 5.1. Each of them shows the combination of the two products that the country can produce with its available resource and technology.

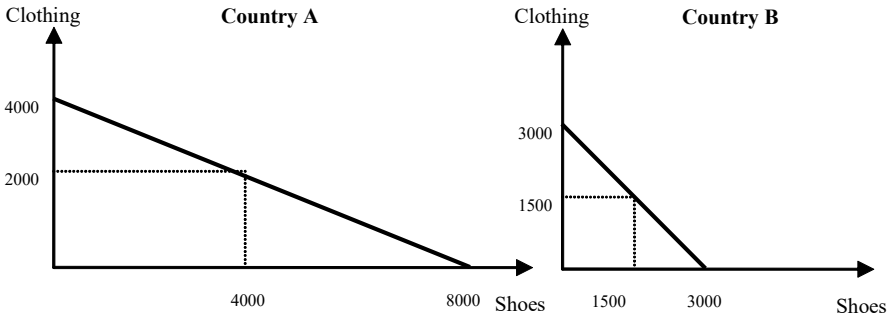


Figure 5.1
Production Possibility Frontiers for Country A and Country B

Case 2: Trade

With trade, each country specialises in the production in which it has comparative advantage. That is, Country A produces shoes only and Country B produces clothing only. However, for ease of comparison with Case 1, we allow Country A to allocate 3,000 units of labour to produce 500 pieces of clothing so that the total number of clothing that will be produced is the same as in Case 1. Table 5.4 shows the production in Country A and Country B with specialisation.

Table 5.4
Production of Country A and Country B with Specialisation

	Production of Shoes	**Production of Clothing**
Country A	21,000/3 = 7,000 pairs	3,000/6 = 500 pieces
Country B	0	24,000/8 = 3,000 pieces
Total	**7,000 pairs**	**3,500 pieces**

Compare Table 5.4 (with specialisation) with Table 5.3 (without specialisation), the total output of shoes is higher after specialisation while the total output of clothing remains unchanged. After specialising in line with their respective comparative advantage, Country A and Country B can get what they want by trading with each other. For example, Country A can trade 2,000 pairs of shoes for 1,500 pieces of clothing with Country B. Table 5.5 describes the quantity of shoes and clothing in Country A and Country B after specialisation and trade.

Table 5.5
Consumption of Country A and Country B with Specialisation and Trade

	Consumption of Shoes	**Consumption of Clothing**
Country A	7,000 – 2,000 = 5,000 pairs	500 + 1,500 = 2,000 pieces
Country B	0 + 2,000 = 2,000 pairs	3,000 – 1,500 = 1,500 pieces

Compare Table 5.5 (with trade) with Table 5.3 (without trade), we find that Country A and Country B end up with 1,000 and 500 pairs of shoes more, respectively, while their quantities of clothing remain unchanged. Therefore, we conclude that both Country A and Country B benefit from specialisation and trade.

It should be noted that to have beneficial trade between the two countries, the terms of trade, that is, the rate at which the two countries exchange the two goods with each other, must lie between the opportunity costs of the two countries. The terms of trade we assume above is 1 pair of shoes for 0.75 piece of clothing (where 0.75 = 1500/2000), which lies between the opportunity cost of Country A (1 pair of shoes for 0.5 piece of clothing) and that of Country A (1 pair of shoes for 1 pair of clothing). As such, both countries can get the good in which they do not have comparative advantage more cheaply by importing it from the other country.

Something to Think About

Sources of Comparative Advantage



According to the Heckscher-Ohlin Theorem (Eli Heckscher and Bertil Ohlin were two Swedish economists in the first half of the 20th century), the existence of a country's comparative advantage is determined by its factor endowments (the types of resources that the country is endowed with, e.g., some countries are endowed with more labour while some countries are endowed with more land). A country has a comparative advantage in the production of a product if that country is endowed more abundantly with inputs used intensively in the production of that product.

What do you think is Hong Kong's comparative advantage? Is your answer consistent with Heckscher-Ohlin Theorem?

Is the promotion of tourism, such as the launching of Mainland Chinese' individual traveller scheme, in line with Hong Kong's comparative advantage? Can you elaborate on your answer?



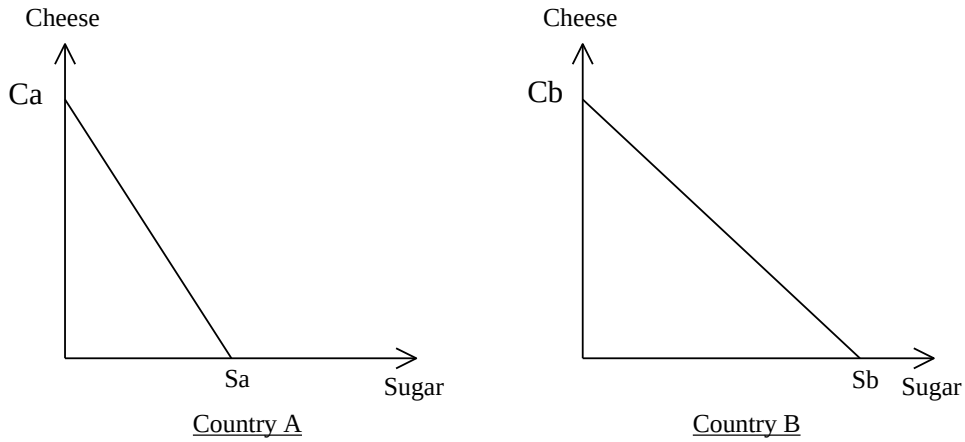
Self Exercise 1

Comparative Advantage and Gain from Trade

There are two countries, namely, Country A and Country B, each of which is endowed with 10 units of resources. Each country produces only cheese and sugar. The productivity of resources in the two countries is as follows.

	Country A		Country B	
	1 tonne of Sugar	1 tonne of cheese	1 tonne of Sugar	1 tonne of cheese
Unit of resources required	2	1	1	1

Consider the following production possibility frontiers (PPF) of the two countries:



Now fill in the blanks in the following questions.

(a) Ca = _____ tonne(s) of cheese.

(b) Sa = _____ tonne(s) of sugar.

(c) Cb = _____ tonne(s) of cheese.

(d) Sb = _____ tonne(s) of sugar.

(e) The opportunity cost of producing 1 tonne of sugar in Country A is _____ tonne(s) of _____ (sugar/cheese).

(f) The opportunity cost of producing 1 tonne of cheese in Country A is _____ tonne(s) of _____ (sugar/cheese).

(g) The opportunity cost of producing 1 tonne of sugar in Country B is _____ tonne(s) of _____ (sugar/cheese).

(h) The opportunity cost of producing 1 tonne of cheese in Country B is _____ tonne(s) of _____ (sugar/cheese).

(i) Country __ (A/B) has an absolute advantage in producing _____ (cheese/ sugar).

(j) Country A has a comparative advantage in producing _____ (cheese/sugar) while Country B has a comparative advantage in producing _____ (cheese/sugar).

Assume that there is no trade between Country A and Country B and that both of them divide their resources equally in the production of cheese and sugar.

(k) The total world output of cheese is _____ tonne(s).

(l) The total world output of sugar is _____ tonne(s).

If each country specializes in the production in which it has a comparative advantage,

(m) the total world output of cheese is _____ tonne(s),

(n) the total world output of sugar is _____ tonne(s),

(o) and the total gain from trade is _____ tonne(s) of cheese plus _____ tonne(s) of sugar.

The response to this self exercise is on page 31.

5.2: Gain from Trade and Trade Barriers

We have concluded in Topic 5.1 with some numerical examples that trade is beneficial to both countries concerned. However, in reality, almost all countries set up various types of trade barriers to regulate the volume of external trade. To explain why it is so common to have trade barriers imposed, we need to use diagrams in this topic. In the first section we will first illustrate the benefits from trade received by the importing and exporting countries of a particular product. Then we will illustrate the redistribution of economic surpluses among different groups of economic agents within each country. The existence of losers in the redistribution is the reason for the trade barriers. In the second section, we will introduce the two most commonly used trade barriers, namely, tariffs and quotas. We will study how each of them works to reduce the volume of trade. In the third section, we will go through the four major arguments for setting up trade barriers.

5.2.1 Gain from Trade

We concluded in Topic 4.5 of Unit 4 that the social surplus would be maximized if the market of a product was perfectly competitive. In other words, there would be no way to increase the social surplus further. This conclusion is true only for a closed economy, one that does not trade with another country. In the real world, however, every country trades with other countries. The increasing importance of international trade among all countries exhibits the phenomenon of globalization. Participating in international trade enables a country to increase its social surplus further, whether it exports or imports a good or service.

To show how a country can benefit from international trade, we assume the small country case in which the trading of the good by this country on the

world market will not have any impact on the world price of the good concerned. In other words, this country's producers and consumers of the good are all price-takers. The world price of the traded good is represented by a horizontal line as shown in Figures 5.2 and 5.3.

Depending on the relationship between the domestic price and the world price of a good, a country will either import or export the good. If the price of a good in the domestic market is higher than the world price, then there will be incentive for its buyers to buy the good at the lower price from the world market. This country will then import the good. On the other hand, if the price of a good in the domestic market is lower than the world price, then there will be incentive for its producers to sell the good at the higher price to overseas buyers in the world market. This country will then export the good. In either case, the domestic price of the good will become the same as the world price after trade.

Figure 5.2 depicts the case of a country that imports cars and shows how the economic surpluses will change after trade. Under autarky, when the country did not take part in international trade, the equilibrium price and quantity of cars were **Pe** and **Qe** respectively. The consumer surplus and producer surplus were given by the areas **AEPe** and **CEPe** respectively, which add up to the social surplus given by area **ACE**. After trade, cars can be imported from the world market at a price of **Pw**, which is below **Pe**. As the domestic price will be lowered to **Pw** after trade, the quantity of cars supplied domestically will drop to **Qs** while the quantity demanded will rise to **Qd**. The domestic shortage of cars will be filled by imported cars as shown in Figure 5.2.

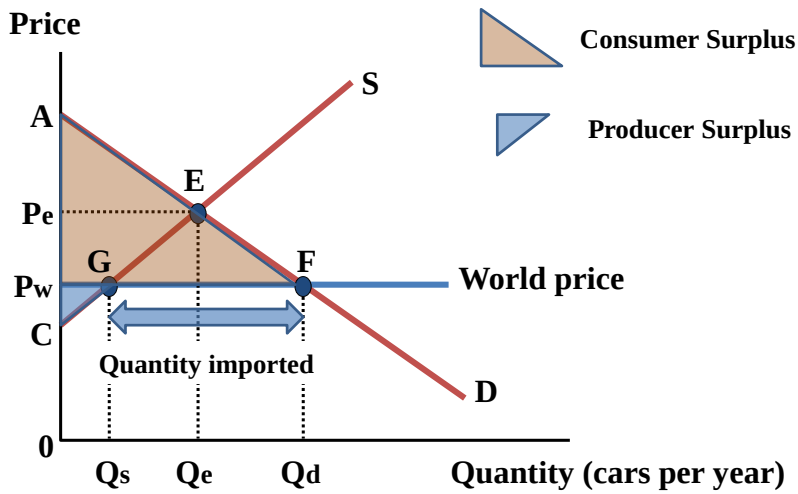


Figure 5.2
Changes in An Importing Country

The resulting lower price and larger quantity of cars consumed will enlarge the consumer surplus to that represented by area **AFPw** while the smaller quantity supplied by domestic car producers at lower price will shrink their

producer surplus to that represented by area **CGPw**. Compared with the social surplus under autarky, the after-trade social surplus will increase by the area **EFG**. Therefore international trade is beneficial to the importing country as a whole.

Figure 5.3 depicts the case of a country that exports T-shirts and shows how the economic surpluses will change after trade. Under autarky, the equilibrium price and quantity of T-shirts as well as the economic surpluses are similar to those explained above with reference to Figure 5.2. After trade, it will be profitable for the domestic producers of T-shirts to export their products to other countries at the world price **Pw**, which is above the domestic price **Pe**. As the domestic price will rise to **Pw** after trade, the quantity of T-shirts supplied by domestic producers will increase to **Qs** while the quantity demanded domestically will drop to **Qd**. The domestic surplus of T-shirts will be absorbed by foreign consumers through exports as shown in Figure 5.3.

The resulting higher price and larger quantity of T-shirts supplied will enlarge the producer surplus to that represented by area **CGPw** while the smaller quantity demanded by domestic consumers at higher price will shrink their consumer surplus to that represented by area **AFPw**. Compared with the social surplus under autarky, the after-trade social surplus will increase by the area **EFG**. Therefore international trade is also beneficial to the exporting country as a whole.

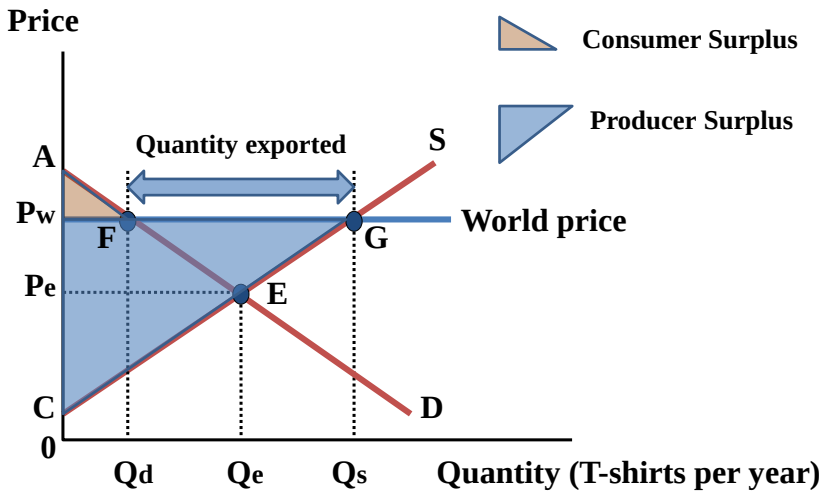


Figure 5.3
Changes in An Exporting Country

5.2.2 Winners and Losers in Free Trade

The above explanation with reference to Figures 5.2 and 5.3 gives us the same conclusion as the analysis of comparative advantage in Topic 5.1 that international trade is beneficial to importing and exporting countries alike.

Then why is it so common to have trade barriers imposed? To answer this question, we have to examine how the economic surpluses of different economic agents are affected in trade.

For an importing country as illustrated in Figure 5.2, while consumer surplus increases with the imports of cars, producer surplus shrinks after trade. In other words, consumers are the winners and producers are the losers. Although the gain to consumers is greater than the loss to producers, which is why the social surplus becomes larger, the losers will not be happy with the unfavourable redistribution of welfare. Conversely, in the case of an exporting country as illustrated in Figure 5.3, producers of T-shirts are the winners but consumers are the losers. In both cases, although engaging in trade is good for the country as a whole, it harms a certain group of agents. Will the losers in both cases resist free trade? The answer is "No".

What we observe in the real world is that only when the losers are producers will there be trade barriers erected to protect their welfare. If consumers are the losers, they tend not to do anything. The explanation is two-fold, both are related to the relatively large number of consumers and much smaller number of producers. First, when the consumers are the losers, due to their large number, the loss per person will be insignificant. When the producers are the losers, due to their small number, the loss per producer or firm will be much more significant, which gives them more incentive to do something to avoid it. Second, when there are a large number of people involved in the fight for some common interest, the individuals tend to behave as free-rider, hoping to benefit from other people's efforts instead of putting in one's own effort. Therefore the consumers tend not to do anything to defend their interest even if they are the losers in trade. On the other hand, the much smaller number of producers makes it much easier for them to work together to defend their welfare. That is why it is very common for the producers to lobby the government to impose trade barriers in an attempt to reduce the quantity of imports that compete with their own products. In the next section, we will examine the trade barriers an importing country may erect to protect its producers.

5.2.3: Different Types of Trade Barriers

The two most commonly used trade barriers are tariffs and quotas.

Tariff

A tariff is a tax on imports. Figure 5.4 illustrates the economic effect of a tariff in a small country case where the importing country is a price-taker.

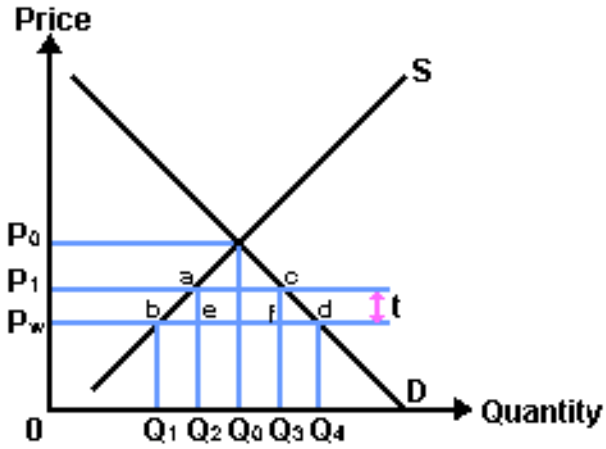


Figure 5.4
Economic Effect of a Tariff (A Small Country Case)

In Figure 5.4, **D** is the demand curve of domestic consumers for the good that can be produced locally or imported whereas **S** is the supply curve of the same good produced by domestic firms. $\mathbf{P_w}$ is the world price of the good. As the importing country is small, its demand and supply of the good have no effect on the world price. $\mathbf{P_w}$ is therefore a horizontal line.

First, let us consider the case under autarky, i.e. without trade. In this case, domestic demand and supply determine the equilibrium. The market equilibrium price is $\mathbf{P_0}$ and the equilibrium quantity is $\mathbf{Q_0}$.

Second, in the case with free trade, the price faced by domestic producers and consumers is the world price, $\mathbf{P_w}$. At $\mathbf{P_w}$, quantity demanded by domestic consumers is $\mathbf{Q_4}$ while quantity supplied by domestic producers is $\mathbf{Q_1}$. The excess demand not satisfied by domestic supply, that is, the quantity $\mathbf{Q_4}$ minus $\mathbf{Q_1}$, will be supplied by foreign producers. Therefore, in the absence of trade barriers, the quantity of imports is $\mathbf{Q_4 - Q_1}$.

If the government of this importing country imposes a tariff of \$**t** on each unit of imports, the price faced by domestic producers and consumers becomes $\mathbf{P_1}$, which is $\mathbf{P_w}$ plus **t**. At $\mathbf{P_1}$, quantity demanded by domestic consumers is $\mathbf{Q_3}$ whereas quantity supplied by domestic producers is $\mathbf{Q_2}$. At this higher domestic price, the excess demand that has to be filled by imports is reduced to $\mathbf{Q_3}$ minus $\mathbf{Q_2}$. Therefore, with the imposition of tariff, the quantity of imports becomes smaller.

How does the tariff-induced reduction in trade volume affect the social welfare of the importing country? We can measure the welfare change using Figure 5.4. There are three parties involved. The welfare loss of consumers is reflected by the reduction in consumer surplus as illustrated by the trapezoid area $\mathbf{P_1cdP_w}$ in Figure 5.4. The welfare gain of domestic producers is reflected by the increase in producer surplus as illustrated by the trapezoid area $\mathbf{P_1abP_w}$. The gain of the government is the tariff revenue

generated, which is equal to $(Q_3 - Q_2) \times t$. Putting these three welfare changes together, we can see that the net welfare loss to the importing country is the sum of the two triangular areas **abe** and **cdf** in Figure 5.4.

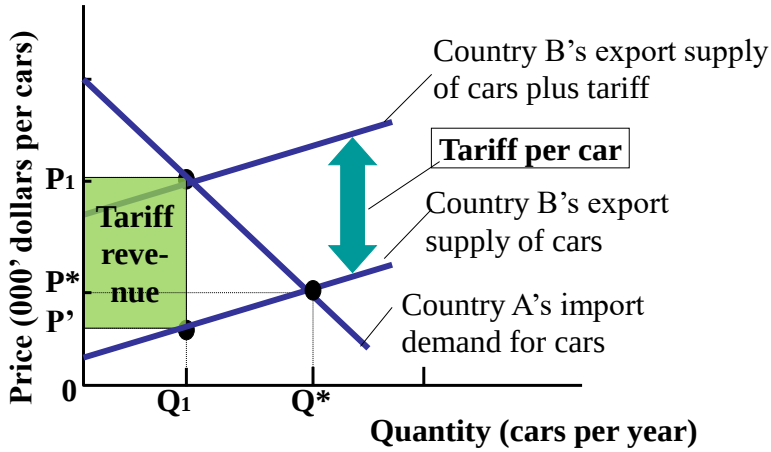


Figure 5.5
Effects of Tariff (A Large Country Case)

Although we have not introduced the large country case in the previous topic for simplicity, as most of the trade barriers involve large countries, we should include Figure 5.5 to illustrate such a case in which the importing country (A) faces an upward-sloping supply curve of the imported goods from Country B. The conclusion about the price, the import volume, and the welfare change are all similar to the small country case analysed above. In essence, the tariff will raise the price from the free-trade level of P* to post-tariff level of P_1. The quantity imported will decrease from Q* to Q_1. The amount of welfare loss will not be discussed in details here.

Quota

A quota is a restriction on the quantity of imports. Figure 5.6 illustrates the economic effect of a quota imposed by the importing country that is again a price-taker.

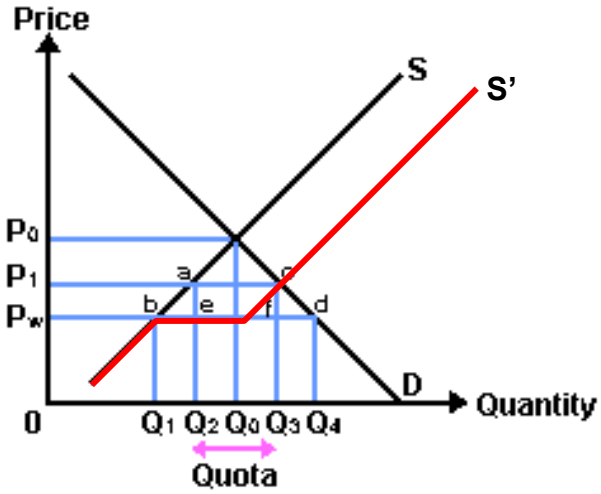


Figure 5.6
Economic Effect of a Quota

Figure 5.6 illustrates the same importing country as does Figure 5.4. The free trade situation is exactly the same as in the previous analysis.

Now, suppose the government of this importing country imposes a quota to restrict the quantity of imports to $\mathbf{Q_3 - Q_2}$. Hence the quota directly reduces the quantity of imports. The curve **S'** in Figure 5.6 depicts the supply curve that includes both the domestic supply and the quota amount. Compared with the free-trade amount of imports, $\mathbf{Q_4 - Q_1}$, this smaller amount of imports will reduce the total supply of the good and drive up the domestic price from $\mathbf{P_w}$ to $\mathbf{P_1}$. At $\mathbf{P_1}$, quantity demanded by domestic consumers is $\mathbf{Q_3}$ and quantity supplied by domestic producers is $\mathbf{Q_2}$.

Figure 5.6 can illustrate the welfare loss due to the imposition of quota. The changes in consumer surplus and producer surplus are the same as in the case of tariff illustrated by Figure 5.4. Unlike tariffs, however, quotas may not yield any revenue for the government. Instead, domestic importers may be able to buy imported goods at $\mathbf{P_w}$ and sell to domestic consumers at $\mathbf{P_1}$. In this case, the profit collected by importers is $(\mathbf{P_1 - P_w})\times(\mathbf{Q_3 - Q_2})$. This amount is known as the **quota rent**.

If the quota rent is captured by importers, the net welfare loss due to the imposition of the quota is the sum of the two triangular areas **abe** and **cdf** in Figure 5.6**,** which is the same as in the case of tariffs. However, sometimes the quota rent may be reaped by the exporters of foreign countries. In that case, the welfare loss from the quota will be larger than in the case of tariff.

Figure 5.7 illustrates the effect of a quota imposed by a large country on its imports of car. Again, the effects are similar to those in the small country case.

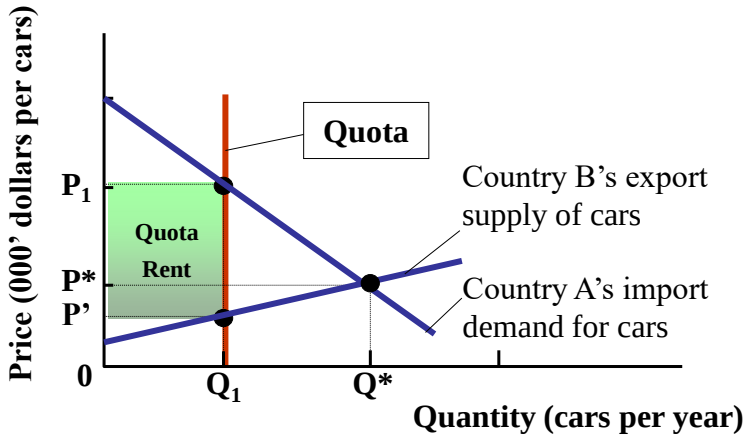


Figure 5.7
Effects of Quota (A Large Country Case)

Self Exercise 2

Impacts of International Trade & Barriers



Suppose the world price of oil is $18 per barrel and a small country X can buy all the oil it wants at the world price. Suppose also that the demand and supply schedules for oil in Country X (in thousands of barrels per day) are as follows:

Price ($ per Barrel)	Quantity Demanded ('000 barrels per day)	Quantity Supplied ('000 barrels per day)
14	16	4
16	15	6
18	14	8
20	13	10
22	12	12

Country X's quantity supplied in the above table refers to the quantity supplied by the domestic oil industry in Country X.

(a) With free trade in oil, what will be the price of oil in Country X? What quantity will the consumers buy? How much of this will be supplied by Country X's producers? How much will be imported?

(b) Suppose Country X imposes a tax of $2 per barrel on imported oil. What quantity will the consumers buy? How much of this will be supplied by domestic producers? How much will be imported?How much tax revenue will the government collect?

The response to this self exercise is on page 33.

Apart from tariff and quota, there are other types of barriers to free trade. Detailed description and analyses are beyond the scope of this subject. They will only be briefly mentioned here. For more details of how they affect the consumers and producers of the traded goods in the countries concerned, please refer to any textbook on the Principles of Economics or International Trade. Please note that the rest of this topic is not required in the examination of AF3625.

Tariff Rate Quota

Tariff rate quota is a hybrid of a quota and a tariff, which applies a lower tariff rate on the imports within the quota and a much higher tariff rate on the imports above the quota.

Voluntary Export Restraints (VER)

Similar to import quota, VER will reduce the quantity of the import of a particular good to the importing country. But unlike quota, VER is imposed by the exporting country at the request or suggestion of the government of the importing country. The quota rent of VER will be captured by the producers in the exporting country.

Local Content Requirements

This is a trade regulation that requires a certain percentage in terms of components or value of an imported good to be produced locally. Such requirements provide protection for domestic producers of the parts of certain goods and harm the consumers of the final products.

Export Subsidies

A government may offer payment to domestic producers of a particular good such as cash grants for exporting, grants for R&D, low-interest loans, or preferential tax treatment, etc. These government supports help lower their prices below the world price and enable them to export their good in the absence of comparative advantage. Note that this is not a barrier to imports but it will distort the resource allocation by encouraging the production of a good that the country is not efficient at.

Administrative Policies

They are bureaucratic rules implemented by a government to make it difficult for foreign goods to enter its country. A typical policy is for the customs inspectors to inspect a large proportion of the imported goods that causes either damage to the goods or delay in their delivery.

5.2.4: Four Arguments for Trade Barriers

In this section we give a brief introduction to four arguments for trade barriers. They are related to the protection of job opportunities, unfair trade practices by trading partners, national security and infant industry. For more

detailed descriptions of these arguments as well as some other arguments not included here, please refer to the textbook.

Protection of Job Opportunities

One argument for trade protection is that it can make consumers buy more domestic goods and thereby create more job opportunities for domestic workers of the protected industry.

However, the consequence of imposing trade barriers is higher prices of imported goods and domestically produced goods of the protected industry. The cost of protection is borne by domestic consumers who have to pay more for the goods.

Unfair Trade Practices by Trading Partners

Some firms export their goods at a price below the cost of production in order to penetrate foreign markets and drive out competitors. This kind of practice is called ***dumping***. Dumping hurts the domestic industries of the importing countries.

To counter the dumping activities of foreign firms and protect domestic industries from this unfair competition, importing countries have to take ***anti-dumping measures***, such as imposing tariffs, to increase the price of imported goods.

National Security

A government is responsible for protecting the nation's key industries that are essential to national defence. Such industries include, for example, the military industry, steel industry and telecommunication industry. National food security is also an argument held by the Chinese government to maintain a high self-sufficiency rate in grain production by means of trade barriers.

Imposing tariffs or quotas can help to maintain the domestic industries and prevent over-dependence on imports of the strategic products.

Infant Industry

This argument states that the government should protect an infant industry that has great potential for development so that it can grow into a matured industry and compete in the world market. Without any protection, an infant industry in a country may have a difficult time competing with established industries in other countries. It is likely to be undercut and driven out of world markets at the beginning of its life.

But economists argue that a more efficient way to nurture an infant industry is for the government to provide subsidies to the producers but not to tax the consumers by imposing tariffs or quotas.

Something to Think About

Trade Distortions in Agriculture



"Food security" is one type of national security that China has been trying to safeguard by imposing trade barriers. The government had set up quotas and tariffs on the imports of grain for many years. Even after China has become a member of WTO since late 2001, it continues to adopt tariff-rate quotas to ensure that more than 90% of rice and wheat are produced locally.

For political reason, many advanced economies including some EU members such as France and Asian countries such as Japan and South Korea have been protecting their farmers by setting up trade barriers on agricultural products. As a result, those developing countries in Africa and South America that have comparative advantage in agricultural products suffer from lower world price of those products.

Using what you have learned from this topic, can you explain how the trade distortions introduced by the protectionism of China and those advanced economies have harmed the welfare of those countries that can produce agricultural products at lower cost?

5.3: Demand and Supply of Foreign Exchange

International trade involves the flow of various currencies in the foreign exchange market. In this topic, we will first define exchange rate. It is the price of one currency in terms of another. Then we will derive the demand curve and the supply curve of foreign exchanges, and discuss the factors that shift the demand curve and supply curve. Finally, we will analyse the determination of exchange rate in two different systems – flexible exchange rate system and fixed exchange rate system.

5.3.1: Exchange Rate

The Exchange rate is the ratio at which two currencies are traded. It is the price of one currency in terms of the other. For example, 14 yen = HK$1, we say that the price of HK$1 is 14 yen, or conversely, the price of 1 yen is HK$0.07. For the sake of consistency, we define the exchange rate as the price of a foreign currency in terms of domestic currency, that is, how many units of domestic currency are required to exchange for one unit of foreign currency. Therefore, the exchange rate between Japanese yen and Hong Kong dollar is follow.

Exchange Rate between Yen and HK$ = HK$/Yen

Appreciation refers to the increase in the exchange value of a currency and ***depreciation*** refers to the decline in the exchange value of a currency.

Before a resident in a country can buy a product made in, or sold by, someone else in another country, a currency swap must take place. Suppose a Hong Kong importer buys a Japanese car. Suppose the yen/HK$ exchange rate is 14 Yen to HK$1, and the yen price of the car is 1.4 million yen, which is HK$100,000. The Hong Kong importer has to exchange HK$100,000 for 1.4 million yen before he/she can import a car from Japan.

The attractiveness of foreign goods to buyers in the domestic country depends in part on the exchange rate. Likewise, the attractiveness of domestic goods to foreign buyers also depends partly on the exchange rate between the currencies of the two countries. If the rate at which yen can be converted to HK$ increases to 17 yen for every HK$ (an appreciation of HK$ against yen, or equivalently, a depreciation of yen against HK$), the price of the same Japanese car falls to HK$82,353 in Hong Kong.

5.3.2: Demand for Foreign Exchange

In this section we will study why the demand curve for a currency is downward sloping and the factors that shift the demand curve for a currency.

Downward Sloping Demand Curve for a Currency

Consider the exchange rate between US$ and HK$. When the exchange value of US$ decreases (the price of US$ in terms of HK$ decreases), the quantity demanded for US$ will increase because Hong Kong people find the US products cheaper in terms of HK$ and they tend to buy more US goods.

Suppose US$ depreciates from US$1:HK$8 to US$1:HK$6, the following table shows what will happen to the demand for US made shoes and the quantity demanded for US$.

Exchange Rate	**US$1 = HK$8**	**US$1 = HK$6**
Price for a pair of US made shoes in US$	US$50	US$50
Price for a pair of US made shoes in HK$	HK$400	HK$300
Quantity demanded for US made shoes	20	28
Quantity demanded for US$	**US$1,000**	**US$1,400**

In the above table, the US$ price for a pair of US made shoes remains unchanged before and after the depreciation of US$, but the HK$ price of a pair of US made shoes falls from HK$400 to HK$300 after the depreciation

of US$. This price falls in terms of HK$ leads to an increase in quantity demanded for US made shoes by the Hong Kong consumers. More imports from the US means larger quantity demanded for US$.

Base on the above analysis, which assumes that the demand for the US made shoes is price elastic, we obtain the following conclusion.

The demand curve for a currency is downward sloping.

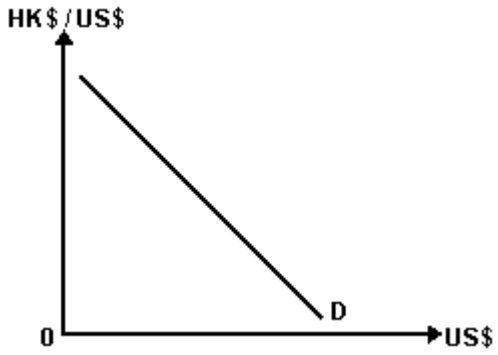


Figure 5.8
Demand Curve for US$

Factors that Shift the Demand Curve for a Currency

There are two major factors that may shift the demand curve for a currency, namely, interest rate and expectation.

- **Interest rate** – If the interest rate in a country increases (decreases) relative to other countries, people will deposit more (less) in that country and therefore the demand for its currency will rise (fall). As a result, the demand curve for the currency of that country will shift to the right (left).

- **Expectation** – If people expect the exchange rate of a currency to increase (decrease) in the future, the demand for that currency will increase (decrease) shifting the demand curve to the right (left).

Expectations about the exchange rate of a currency are formed on the basis of the expectations of the price level and the interest rate in that country. For instance, if investors expect the price level in a particular country to increase in the near future, they will expect the exchange rate of its currency to depreciate because a higher price level makes this country's exports less competitive in the world market.

As the interest rate of a currency has a positive impact on the currency's exchange value, an expected increase in interest rate in a country will create an expectation of appreciation of its currency.

5.3.3: Supply of Foreign Exchange

In this section we will explain why the supply curve of a currency is upward sloping and study the factors that shift the supply curve of a currency.

Upward Sloping Supply Curve of a Currency

Consider the exchange rate between US$ and HK$. When the exchange value of US$ decreases (the price of US$ in terms of HK$ decreases), quantity supplied of US$ will decrease because the US citizens find the Hong Kong products more expensive in terms of US$ and thus buy less Hong Kong goods.

Suppose US$ depreciates from US$1:HK$8 to US$1:HK$6, the following table shows what will happen to the demand for Hong Kong made shoes and the quantity supplied of US$.

Exchange Rate	**US$1 = HK$8**	**US$1 = HK$6**
Price for a pair of HK made shoes in HK$	HK$120	HK$120
Price for a pair of HK made shoes in US$	US$15	US$20
Quantity demanded for HK made shoes	30	22
Quantity supplied of US$	**US$450**	**US$440**

In the above table, even though the HK$ price of a pair of HK made shoes remains unchanged before and after the depreciation of US$, the US$ price of a pair of HK made shoes rises from US$15 to US$20 after the depreciation of US$. This rise in US$ price level leads to a decrease in quantity demanded for HK made shoes by the US residents. Less imports by the US results in a decrease in quantity supplied of US$ from US$450 to US$440.

Based on the above analysis, which assumes that the demand for Hong Kong made shoes is price elastic, we obtain the following conclusion.

The supply curve of a currency is upward sloping

Figure 5.9 is a supply curve of US$.

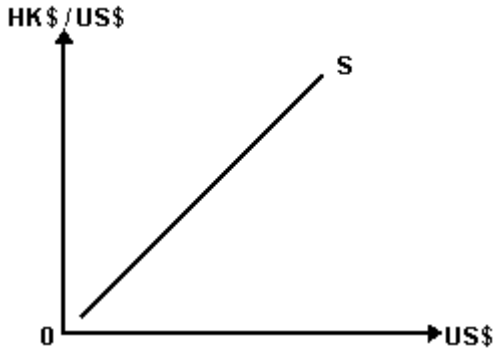


Figure 5.9
Supply Curve of US$

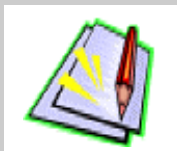
Factors that Shift the Supply Curve of a Foreign Currency

Similar to the demand curve of a currency, there are two major factors that may shift the supply curve of a currency, namely, interest rate and expectation.

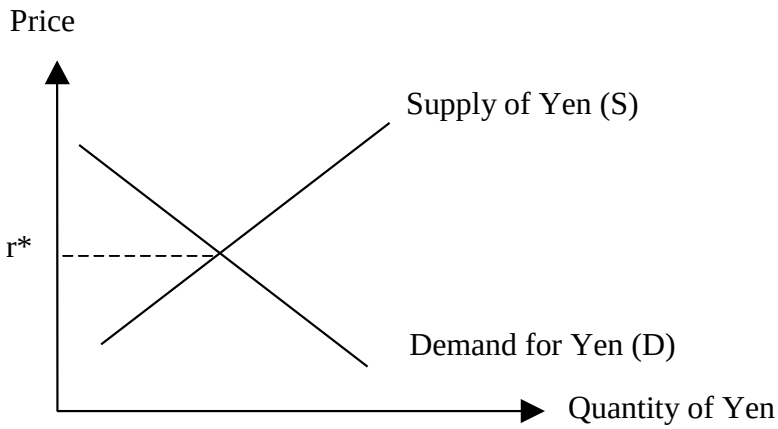
- **Interest rate** – If the interest rate in a country increases (decreases) relative to other countries, foreign investors will deposit more (less) in that country and therefore the supply curve of foreign currency will shift to the right (left).
- **Expectation** – If people expect the exchange rate of the domestic currency to increase in the future, they will sell foreign currency in exchange for this particular currency. As a result, the supply curve of foreign currency will shift to the right. Conversely, if people expect the exchange rate of the domestic currency to fall in the future, the supply curve of foreign currency will shift to the left.

Self Exercise 3

Determinants of Demand and Supply of a Currency



Consider a transaction between Japan and the US. The following figure represents the foreign exchange market for yen and US$.



Now complete the following sentences.

(a) The price for yen should be expressed in terms of _____ ($\frac{Yen}{US\$}, \frac{US\$}{Yen}$).

(b) A movement up the vertical axis represents _____ (a depreciation / an appreciation) of the dollars and _____ (a depreciation / an appreciation) of yen.

(c) An increase in Japanese exports to the US will shift the (demand/supply) curve for yen to the (left/right) while an increase in Japanese imports from the US will shift the (demand/supply) curve for yen to the (left/ right).

(d) An increase in American exports to Japan, all other things being equal, will cause (a depreciation/ an appreciation) of the dollars while an increase in American imports from Japan will cause (a depreciation/ an appreciation) of yen.

Suppose that the exchange market is currently in equilibrium at the exchange rate r*. If the interest rates on the US government bonds increase relative to the interest rates on Japanese government bond.

(e) The demand for yen will _____ (shift to the right / shift to the left/ remain constant) and the supply of yen will _____ (shift to the right / shift to the left/ remain constant).

(f) At the original exchange rate r* there will now be an excess _____ (demand for/ supply of) yen.

(g) The new equilibrium exchange rate will be _____ (higher/ lower) than r*.

The response to this self exercise is on page 34.

5.4: Determination of Exchange Rate

Having discussed the determinants of demand and supply of foreign exchanges, we are now ready to study the equilibrium of foreign exchange market.

This topic will be divided into three sections. The first section examines the determination of equilibrium exchange rate of a currency under a flexible exchange rate system. The second section examines the fixed exchange rate system and how a government can maintain the exchange rate of its currency at a target level. The third section gives a brief description of the linked exchange rate system of the Hong Kong dollar.

5.4.1: Flexible Exchange Rate System

Under a flexible exchange rate system, the equilibrium exchange rate is determined by the forces of demand and supply without government intervention.

If demand for foreign currency increases due to, for example, an increase in imports, foreign currency will become more expensive (appreciation of foreign currency).

If supply of foreign currency increases due to, for example, an increase in exports, foreign currency will become less expensive (depreciation of foreign currency).

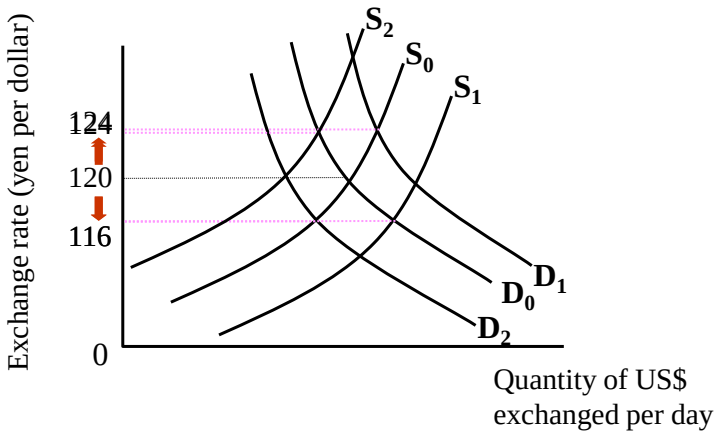


Figure 5.10
Equilibrium under a Flexible Exchange Rate System

Take yen and US$ as an example. Figure 5.10 illustrates an original equilibrium exchange rate at US$1:120 yen, which is the intersection of the demand curve and supply curve of US$, denoted by $\mathbf{D_0}$ and $\mathbf{S_0}$ respectively.

An increase in demand for US$, as illustrated by the shift from $\mathbf{D_0}$ to $\mathbf{D_1}$, with supply unchanged will increase the equilibrium exchange rate (from 120 to 124 yen/US$). Conversely, a decrease in demand for US$, as illustrated by the shift from $\mathbf{D_0}$ to $\mathbf{D_2}$, with supply unchanged will decrease the equilibrium exchange rate (from 120 to 116 yen/US$).

Similarly, a change in supply of US$ with demand unchanged will also lead to an adjustment in the equilibrium exchange rate, as illustrated by the shifts in the supply curve from $\mathbf{S_0}$ to $\mathbf{S_1}$ and $\mathbf{S_2}$ in Figure 5.10.

5.4.2: Fixed Exchange Rate System

While most developed countries adopt flexible exchange rate system, many developing countries adopt fixed exchange rate system for their currencies.

China is one of them. Let us explain how a fixed exchange rate system operates using the Chinese currency, renminbi (RMB or yuan), as an example.

Since the unification of the official exchange rate and the unofficial swap rate of RMB in 1994, the exchange rate of the Chinese currency had been fixed within a very narrow range, as can be seen in the following table. Even after the eruption of the Asian financial crisis in 1997, which prompted many Asian countries to devalue their currencies, the Chinese government maintained the fixed rate of RMB against US$.

Table 5.6
The Exchange Rate of RMB against US$, Japanese Yen and Hong Kong Dollar, 1997-2001

Year	US$100	100 yen	HK$100
1997	828.98	6.8600	107.09
1998	827.91	6.3488	106.88
1999	827.83	7.2932	106.66
2000	827.84	7.6864	106.18
2001	827.70	6.8075	106.08

Source: China Statistical Yearbook 2002

As China opens up to international trade, it has been a net exporter of commodities. At the same time, China has attracted a large amount of foreign direct investment (FDI) making it the world's second largest FDI recipient only after the United States. Based on our previous analyses of the demand and supply of a currency, it is obvious that the international trade and the capital movements in recent years will have increased the demand for RMB and increased the supply of foreign currencies. It must be noted that the two are equivalent as they are meant to be exchanged for one another.

To analyse the exchange rate of a domestic currency, in this case the RMB, it is a common practice to depict the demand and supply of a foreign currency. In the following analysis, we study the exchange rate of RMB against US$. Figure 5.9 shows the demand and supply of US$ while the exchange rate is the price of US$ in terms of yuan. $\mathbf{D_0}$ and $\mathbf{S_0}$ denote the original demand for and supply of US$ in exchange for RMB. Assume that they give an equilibrium exchange rate of 8.27 yuan/US$, which is the same as the target rate set by the Chinese government.

Note that China has adopted a more flexible exchange rate policy since 2005 and RMB has appreciated against US$ substantially since then. For simplicity, we assume the same exchange rate that prevailed before 2005 in the analyses of the following two cases.

Case 1: An increase in supply of US$

Given China's trade surpluses the inflow of capital due to FDI, there will be an increase in the supply of foreign currency. The supply curve of US$ will shift to the right from $\mathbf{S_0}$ to $\mathbf{S_1}$ in Figure 5.9 creating a downward pressure on the equilibrium exchange rate of US$ (i.e. a pressure for RMB to appreciate). To maintain the target exchange rate at 8.27 yuan/US$, the Chinese government has to buy US$ using RMB. Such an intervention in the foreign exchange market is represented by an increase in demand for US$. The resulted shift in the demand curve from $\mathbf{D_0}$ to $\mathbf{D_1}$ in Figure 5.11 will restore the equilibrium exchange rate at the target level. In so doing, the Chinese government accumulates foreign exchange reserves.

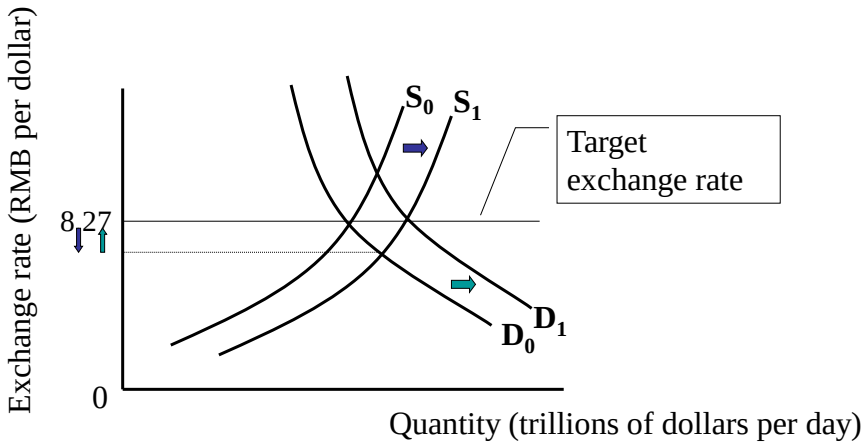


Figure 5.11
Market Intervention to Maintain Target Exchange Rate
when Supply of Foreign Exchange Increases

Case 2: An increase in demand for US$

Suppose China's balance of payments situation had reversed and that China had been importing more from the US than exporting to it. The change would result in an increase in demand for US$, as illustrated by the shift in demand curve from $\mathbf{D_0}$ to $\mathbf{D_1}$ in Figure 5.10. Such a change will create a depreciation pressure on RMB. In order to keep the exchange rate at the target rate of 8.27 yuan/US$, the Chinese government would have to sell US$ for RMB. Such an intervention in the foreign exchange market is represented by an increase in supply of US$. The resulted shift in the supply curve from $\mathbf{S_0}$ to $\mathbf{S_1}$ in Figure 5.12 will restore the equilibrium exchange rate at the target level. In so doing, the Chinese government loses foreign exchange reserves.

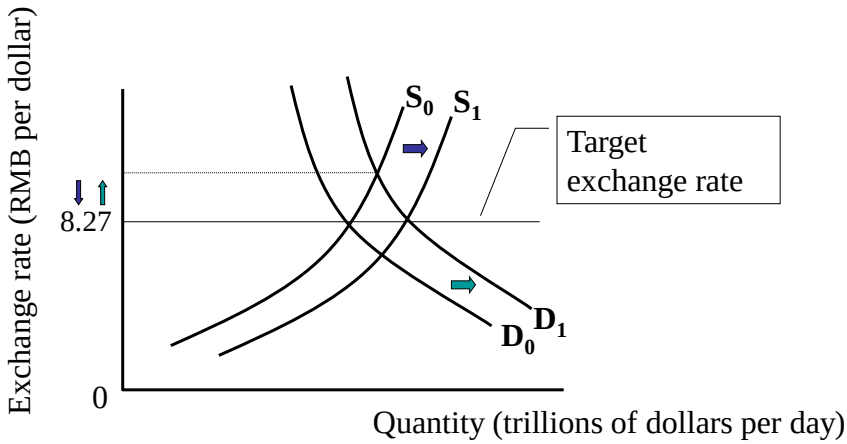


Figure 5.12
Market Intervention to Maintain Target Exchange Rate
when Demand for Foreign Exchange Increases

It should be noted that persistent intervention in the foreign exchange market cannot be sustained. For example, if a government has to keep selling foreign exchange reserves to maintain the target exchange rate of its domestic currency that is constantly under depreciation pressure, it will deplete its official reserves sooner or later. When the government gives up the attempt to maintain an overvalued domestic currency, it may allow it to devalue. ***Devaluation*** is a downward adjustment in the fixed exchange value of the domestic currency. On the other hand, ***revaluation*** is an upward adjustment in the fixed exchange value of the domestic currency.

5.4.3: The Linked Exchange Rate System of the Hong Kong Dollar

Since 17 October 1983, the Hong Kong dollar has been pegged to US$ at a rate of HK$7.8 to US$1. The purpose of the linked rate system was to stabilise the value of HK$ amid the confidence crisis stemming from political uncertainties about the future of Hong Kong. Different from a fixed exchange rate system that is maintained through government intervention in the foreign exchange market, the maintenance of the exchange rate of HK$7.8 to US$1 is through an automatic mechanism that adjusts Hong Kong's money supply according to the capital flows of foreign currency.

The basic mechanism of the linked exchange rate system in Hong Kong is a currency board system, which requires full backing of both the stock and flow of the monetary base by foreign reserves. To understand how it works, we have to first study the note-issuing mechanism under this system.

When the three note-issuing banks in Hong Kong issue banknotes, they are required by law to purchase certificates of indebtedness (CIs) from the Hong Kong Monetary Authority. For example, if the Bank of China is to

issue HK$780 of bank notes, it has to submit US$100 to the HKMA in exchange for CIs of HK$780. When the bank wants to withdraw a certain amount of banknotes from circulation, it surrenders CIs to HKMA and receives an equivalent amount of US$. The HKMA stands ready to buy and sell Hong Kong dollar from and to banks at the convertibility rate of HK$7.8 to US$1.

How does this currency board system maintain the stability of the linked exchange rate? It is through an automatic interest rate adjustment mechanism. When there is a decrease in demand for Hong Kong dollar assets (i.e. a capital outflow), it will weaken the Hong Kong dollar such that the market exchange rate will be higher than HK$7.8/US$. Banks will be induced to arbitrage between the market rate and the linked rate by buying HK$ from the foreign exchange market and selling it to the HKMA. As the HKMA stands ready to purchase Hong Kong dollars (or redeem CIs) from banks, it leads to a contraction of the monetary base and hence the money supply. Interest rates then rise and attract capital inflows. These inflows will offset the original outflow and maintain exchange rate stability.

Conversely, if there is an increase in the demand for Hong Kong dollar assets (i.e. capital inflow), leading to a strengthening of the exchange rate (i.e. the market exchange rate falls below HK$7.8/US$), banks will be induced to sell HK$ on the foreign exchange market and buy it from the HKMA. As HKMA stands ready to sell Hong Kong dollars at the linked rate to the banks (or issue CIs required for issuing Hong Kong dollar notes), the monetary base expands, which in turn increases the money supply. Interest rates then fall and discourage continued inflows of capital.

For more detailed information of Hong Kong's linked exchange rate system, visit the following website of HKMA.

http://www.hkma.gov.hk/eng/key-functions/monetary-stability/linked-exchange-rate-system.shtml

Self Exercise 4

Impact of Government Policies
on Exchange Rate



	Government Policy	Links with		Impact
1.	If a country fixes its exchange rate above the equilibrium level,	●	●	a. it automatically adjusts to eliminate balance of payments surplus and deficit.
2.	When a government is operating with a fixed exchange rate and the currency is undervalued,	●	●	b. its currency will tend to appreciate.
3.	The main argument proposed by economists for flexible exchange rate is that	●	●	c. it will have to buy domestic currency and sell foreign exchange.
4.	Under a flexible exchange rate system, if a country's inflation rate decreases relative to inflation rates elsewhere,	●	●	d. its foreign exchange reserves will keep accumulating.

The response to this self exercise is on page 35.

Must Read Before Moving On …

Now we have finished the first five units, i.e. the first four units on microeconomics and the fifth unit on macroeconomics, which comprise the first part of this subject.

Throughout these five units, we have tried to present a balanced description of how an economy functions, both in theory and in practice. Principles of economics is an interesting subject because it helps us understand all sorts of phenomena in our daily lives, which are the outcomes of millions of economic decisions, from buying a candy to building a factory. As in the previous four units, please check through the list of objectives to make sure that you have achieved all of them before moving on to the second part of this subject.

You will have achieved one of the most important objectives of this course if now you find yourself more able to understand the economic news on the newspapers and other mass media. Once you are equipped with the economic concepts and analytical skills, you can analyse whether an economic argument put forward by a columnist, an economist, or a politician is justified. Hopefully you have not been put off by the abstract economic concepts and the applications of theory to real-life problems have aroused your interests in real world economic phenomena.

In the next two units you will do something less abstract and more closely related to your future career as an engineer. All the analyses will mostly be conducted in mathematics, which may be easier for you.

Responses to Self Exercises

This section contains the responses to the four (4) self exercises in unit 5 – International Trade and Finance. I believe you can get the most from this course if you can attempt the self exercises immediately after you have covered the related topics and check your understanding. The additional information contained in the responses will further enhance your understanding of the topic concerned.

Response to Self Exercise 1

Comparative Advantage and Gain from Trade

The answers are given below.

(a) Ca = **<u>10</u>** tonne(s) of cheese.

Explanation: It is the vertical intercept on Country A's PPF, which gives the maximum amount of cheese that can be produced with all the resources of Country A.

(b) Sa = **5** tonne(s) of sugar.

Explanation: It is the horizontal intercept on Country A's PPF, which gives the maximum amount of sugar that can be produced with all the resources of Country A.

(c) Cb = **10** tonne(s) of cheese.

Explanation: It is the vertical intercept on Country B's PPF, which gives the maximum amount of cheese that can be produced with all the resources of Country B.

(d) Sb = **10** tonne(s) of sugar.

Explanation: It is the horizontal intercept on Country B's PPF, which gives the maximum amount of sugar that can be produced with all the resources of Country B.

(e) The opportunity cost of producing 1 tonne of sugar in Country A is **2** tonne(s) of **cheese**.

(f) The opportunity cost of producing 1 tonne of cheese in Country A is **1/2** tonne(s) of **sugar**.

(g) The opportunity cost of producing 1 tonne of sugar in Country B is **1** tonne(s) of **cheese**.

(h) The opportunity cost of producing 1 tonne of cheese in Country B is **1** tonne(s) of **sugar**.

(i) **Country B** has absolute advantage in producing **sugar**.

Explanation: Country B is more efficient in producing sugar. In other words, Country B needs fewer resources than Country A does in order to produce 1 tonne of sugar. In addition, Country A and Country B are equally efficient in producing cheese.

(j) Country A has a comparative advantage in producing **cheese** while Country B has a comparative advantage in producing **sugar**.

Explanation: One common mistake is the confusion of absolute advantage with comparative advantage. Unlike absolute advantage, comparative advantage is measured in terms of opportunity cost. From the answers to part a - h above, one can see that the opportunity cost of producing cheese is lower in Country A while the opportunity cost of producing sugar is lower in Country B.

Assume that there is no trade between Country A and Country B and that both of them divide their resources equally in the production of cheese and sugar.

(k) The total world output of cheese is **10** tonne(s).

Explanation: Each of the two countries uses 5 units of resources to produce cheese. As a result, 5 + 5 = 10.

(l) The total world output of sugar is **7.5** tonne(s).

Explanation: Each of the two countries uses 5 units of resources to produce sugar. As a result, 5/2 + 5 = 7.5.

If each country specializes in the production in which it has comparative advantage,

(m) the total world output of cheese is **10** tonne(s);

Explanation: From the answers to part j, we know that Country A specializes in cheese. Consequently, it produces 10 units of cheese.

(n) the total world output of sugar is **10** tonne(s);

Explanation: From the answers to part j, we know that Country B specializes in sugar. Consequently, it produces 10 units of sugar.

(o) and the total gain from trade is **0** tonne(s) of cheese plus **2.5** tonne(s) of sugar.

Explanation: The answer can be obtained simply by comparing the answers to m & n with those to k and l. This question illustrates why free trade is beneficial to the countries involved.

Response to Self Exercise 2

Impacts of International Trade & Barriers

This question should be very straightforward. You can apply the demand-supply analysis to derive the equilibrium easily. Here are the answers.

(a) After trade the domestic price will be the same as the world price at $18. At this price the consumers of Country X will demand 14,000 barrels of oil per day but domestic producers will be willing to supply only 8,000 barrels. Country X has to import 6,000 barrels of oil per day to fill the shortage.

(b) With the $2 tariff imposed, the domestic price or oil becomes $20. At this price the consumers of Country X will demand 13,000 barrels of oil per day but domestic producers will be willing to supply only 10,000 barrels. Country X has to import 3,000 barrels of oil per day to fill the shortage. Total tax revenue generated by the tariff for the government will be $6,000.

Response to Self Exercise 3

Determinants of Demand and Supply of a Currency

The answers and explanations are listed below.

(a) The Price for yen on the vertical axis should be expressed in terms of $\frac{\textbf{\textit{US\$}}}{\textbf{\textit{Yen}}}$.

Explanation: That is the price of yen in terms of US$.

(b) A movement up the vertical axis represents **a depreciation** of the dollars and **an appreciation** of yen.

Explanation: More US$ is needed to exchange for 1 yen. That implies an increase in the price of yen in terms of US$.

(c) An increase in Japanese export to the US will shift the **demand** curve for yen to the **right** while an increase in Japanese imports from the US will shift the **supply** curve for yen to the **right**.

Explanation: An increase in exports means that foreign importers need more local currency to buy local goods and services. An increase in imports means that domestic importers need to sell more local currency in exchange for foreign currency to pay for imported goods and services.

(d) An increase in American exports to Japan, all other things being equal, will cause **an appreciation** of the dollars while an increase in American imports from Japan will cause **an appreciation** of yen.

Explanation: The question is a continuation of (c). It is simply an adjustment in equilibrium exchange rate following the shift in demand or supply of the currencies concerned.

(e) The demand for yen will **shift to the left** and the supply of yen will **shift to the right**.

Explanation: The US government bonds will be relatively more attractive so there will be a decrease in demand for yen (an increase in demand for US$). Some of you may be able to identify the change in demand but not the change in supply. As the investors of Japanese bonds will find their investment less attractive than the US government bonds, they will increase their supply of yen after selling Japanese bonds and then purchase US dollars in order to purchases the US government bonds.

(f) At the original exchange rate r* there will now be an excess **supply of** yen.

Explanation: Following from (e), the quantity supplied of yen is greater than the quantity demanded for yen at the original exchange rate after the movements of demand and supply curves.

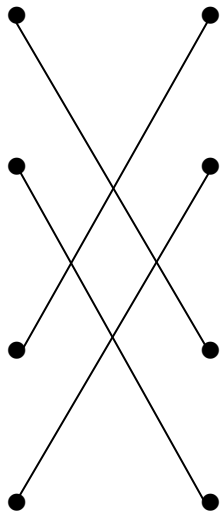
(g) The new equilibrium exchange rate will be **lower** than r*.

Explanation: Following from (f), the excess supply of yen will create a downward pressure on the exchange rate of yen.

Response to Self Exercise 4

Impact of Government Policies on Exchange Rate

Government Policy	Links with	Impact
1. If a country fixes its exchange rate above the equilibrium level,		a. it automatically adjusts to eliminate balance of payments surplus and deficit.
2. When a government is operating with a fixed exchange rate and the currency is undervalued,		b. its currency will tend to appreciate.
3. The main argument proposed by economists for flexible exchange rate is that		c. it will have to buy domestic currency and sell foreign exchange.
4. Under a flexible exchange rate system, if a country's inflation rate decreases relative to inflation rates elsewhere,		d. its foreign exchange reserves will keep accumulating.



Excellent if you managed to match them all correctly! You should be pleased with yourself if you have got all of them right.

This exercise focuses on the differences between the two systems of exchange rate, namely, the fixed exchange rate and the flexible exchange rate.

If the government implements a flexible exchange rate system, any excess demand and excess supply in the foreign exchange market can be eliminated by appreciation and depreciation, respectively. Note that in the above exercise, the term "undervalued" occurs when there is a balance of payments surplus.

If the government implements a fixed exchange rate system, the government has to "fill up" the excess demand or excess supply of domestic currency by buying or selling foreign exchange reserves, respectively.

These materials were produced by Dr. Yi Ping Shea. Parts of the contents were based on an earlier version produced by Dr. Mike Fung for Unit 12 of the Self Study Notes for Introduction to Economics.

This is the end of Unit 5.